

Images

An Autonomous Neuromorphic Underwater Coral Reef Restoration System with Event Based Detection and Adaptive Indexed Pod Deployment

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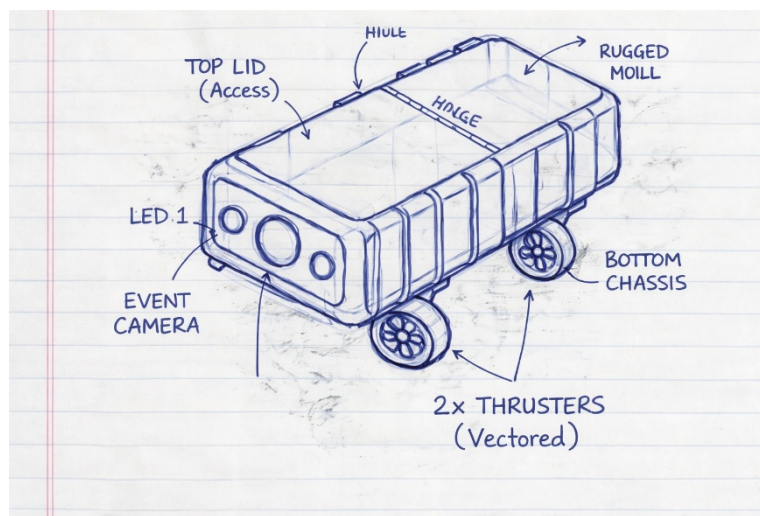


Figure 1: A rugged coral restoration device featuring a top access hinged lid, front mounted event driven camera with dual LEDs, and vectored thrusters for stable navigation and deployment.

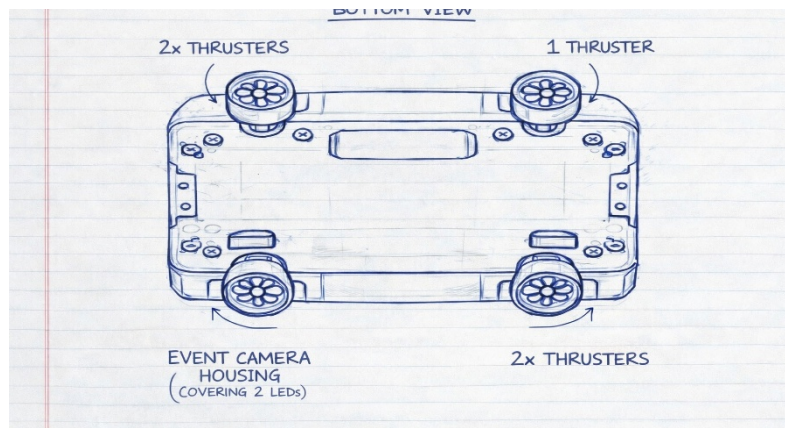


Figure 2: Bottom view showing four vectored thrusters and a protected event camera housing with dual LED coverage for underwater stability and vision.

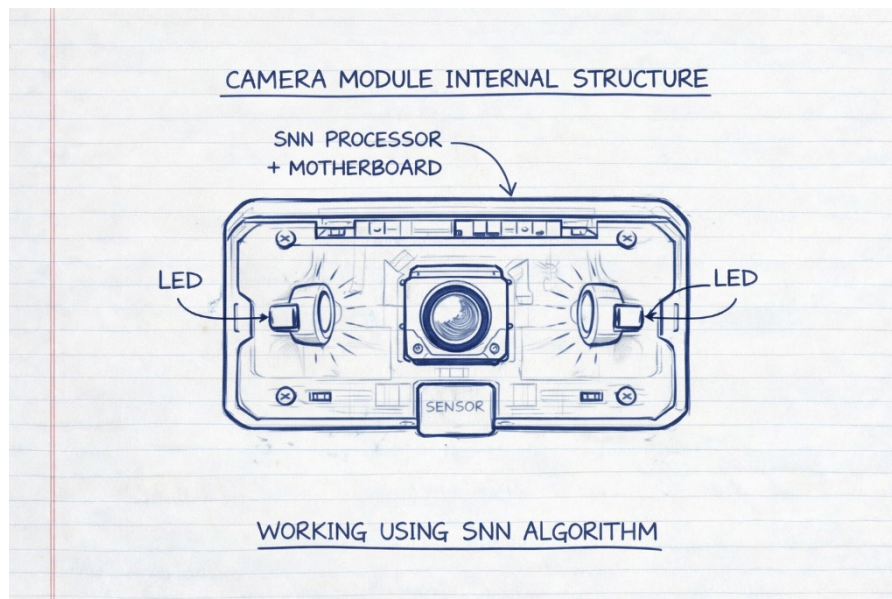


Figure 3: Internal camera module featuring a central event driven sensor with dual LED's and an integrated SNN processor for real time underwater reef analysis.

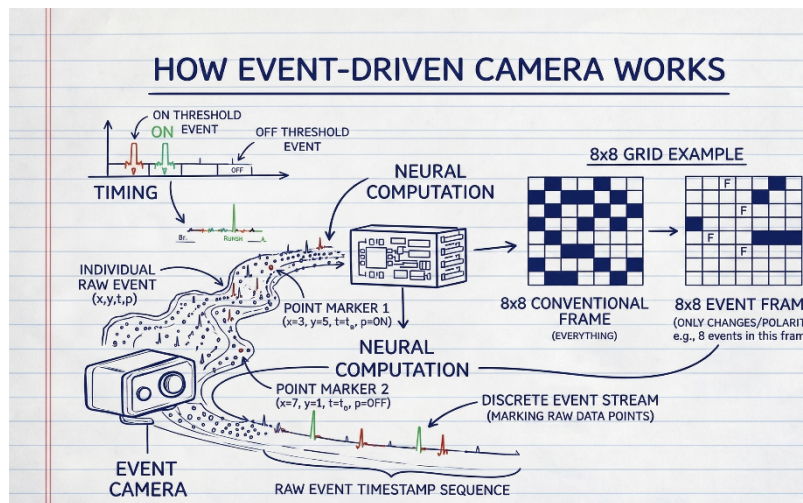


Figure 4: An event driven camera captures only changes in brightness at each pixel and sends them as fast, spike like signals instead of recording full image frames continuously.

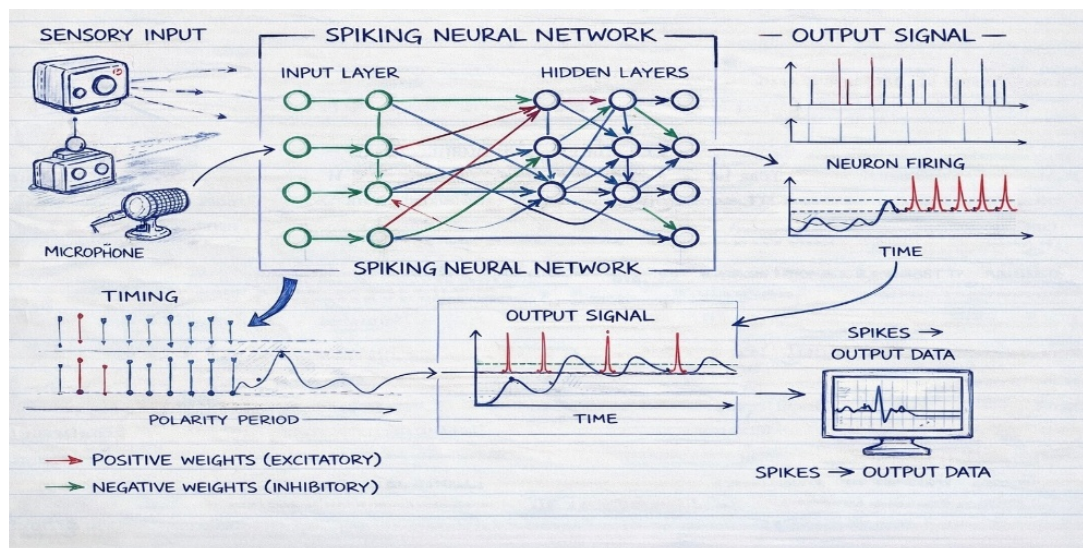


Figure 5: A Spiking Neural Network (SNN) processed the time based sensory inputs as discrete spikes, using excitatory and inhibitory neuron firing pattern to generate efficient and event driven output signals.

HOW X-Y ELECTRIC GANTRY/ INDEXING SYSTEM WORKS

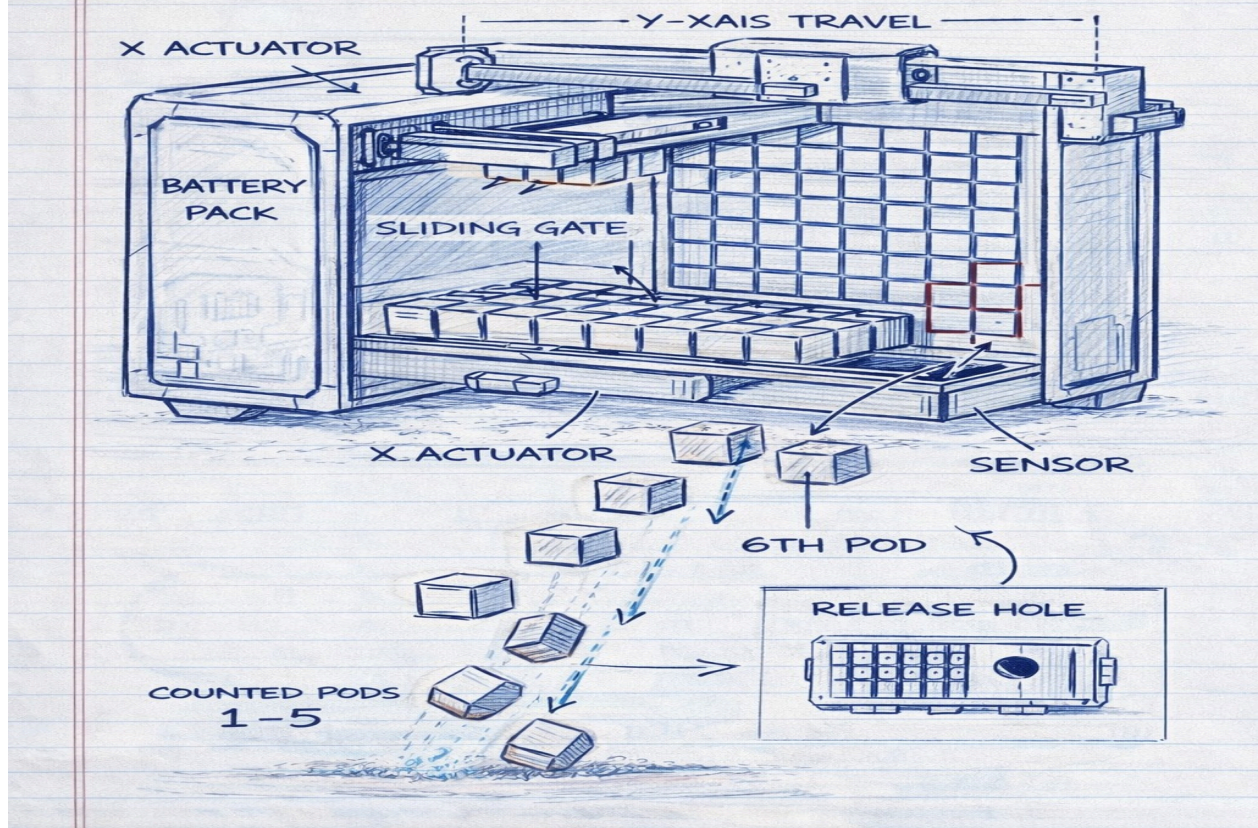


Figure 6: The XY electric gantry indexing system precisely moves across a pod grid using actuators and sensors to align and release selected coral pods through a controlled deployment hole.

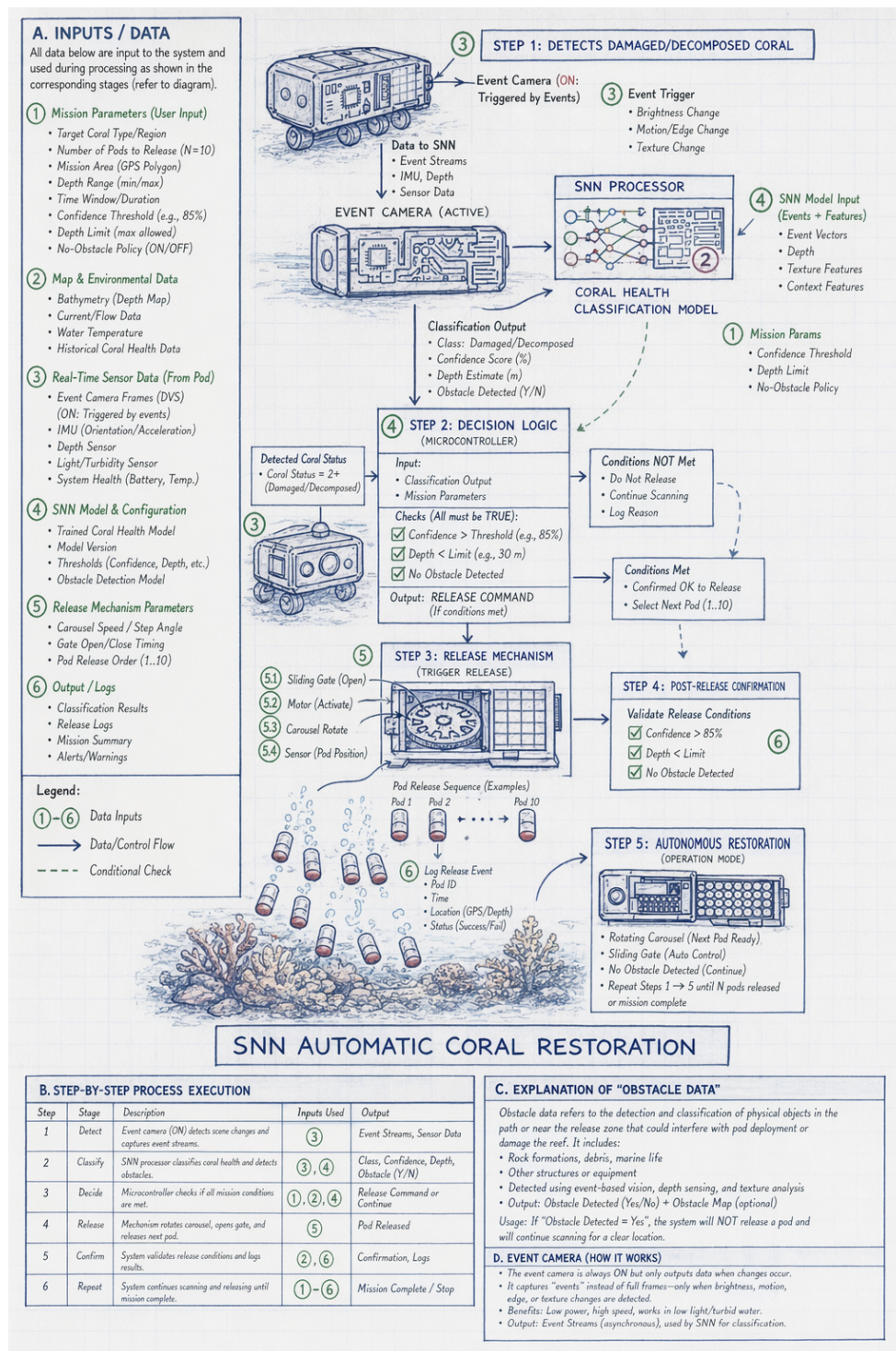


Figure 7 : An SNN based autonomous system detects damaged coral, validates environmental conditions and automatically deploys biodegradable pods through a controlled release mechanism for reef restoration.